

University of Bedfordshire

BSc (Hons) Software Engineering

Nepal Advocate

A Comprehensive Mobile Platform for Digitizing and
Democratizing Legal Access in Nepal

Research Methodology and Emerging Technology — Contextual Report

Submitted by: Sujal Kunwar | University ID: 2337702

Supervisor: Pawan Kc | Course Coordinator: Ajaya Kumar Sharma

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ABSTRACT

The legal information ecosystem in Nepal is characterized by fragmentation, outdated digital resources, and significant accessibility barriers for citizens, students, and legal practitioners outside the Kathmandu Valley. Existing digital solutions, such as the official Nepal Law Commission application and the Supreme Court's Nepal Kanun Patrika (NKP) portal, suffer from poor user interface design, limited search capabilities especially for Devanagari Unicode text, and a lack of offline functionality necessary for a country with inconsistent internet connectivity.

This report proposes the design and development of "Nepal Advocate," a unified mobile application intended to bridge this access-to-justice gap. The proposed system architecture integrates a React Native frontend, a Firebase backend, and a typo-tolerant search engine (Algolia) specifically tuned for Nepali Unicode. Key features include a verified lawyer directory with Bar Council license validation, offline-accessible Bare Acts, a searchable database of Supreme Court precedents, and an integrated payment system for booking consultations.

This document provides a comprehensive feasibility study, detailed system design, implementation methodology, and a thorough analysis of the legal and ethical considerations required to bring this platform to market. The report also introduces an AI integration roadmap featuring base model tuning of Small Language Models (SLMs) on Nepali legal corpora for Retrieval-Augmented Generation (RAG). The report concludes that the Nepal Advocate app is technically viable and socially imperative, offering a sustainable model for modernizing legal access in Nepal.

Keywords: Legal Technology, Mobile Application, Access to Justice, Devanagari Unicode, React Native, Firebase, Nepal Legal System, AI Legal Assistant, Base Model Tuning, RAG

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LIST OF ABBREVIATIONS

NKP:Nepal Kanun Patrika (Official Law Report)
BS:Bikram Sambat (Nepali Calendar)
API:Application Programming Interface
UI/UX:User Interface / User Experience
TAM:Technology Acceptance Model
SQL:Structured Query Language
NoSQL:Not Only SQL
JSON:JavaScript Object Notation
PDF:Portable Document Format
SDK:Software Development Kit
PCI-DSS:Payment Card Industry Data Security Standard
TLS:Transport Layer Security
AES:Advanced Encryption Standard
UAT:User Acceptance Testing
WCAG:Web Content Accessibility Guidelines
NTA:Nepal Telecommunications Authority
RAG:Retrieval-Augmented Generation
SLM:Small Language Model
LLM:Large Language Model
NLP:Natural Language Processing
LoRA:Low-Rank Adaptation
PEFT:Parameter-Efficient Fine-Tuning

CHAPTER 1: INTRODUCTION

1.1 Background

The legal framework of the Federal Democratic Republic of Nepal is a complex tapestry woven from the Constitution of Nepal 2072 (2015), the National Civil and Criminal Codes of 2074 (2017), and decades of judicial precedents documented in the Nepal Kanun Patrika (NKP). Access to these fundamental legal resources is a constitutional right and a practical necessity for legal professionals, students, and the general citizenry. However, the digitization and dissemination of these resources remain fragmented and inefficient.

Currently, primary access points are limited to the official websites of the Nepal Law Commission (lawcommission.gov.np) and the Supreme Court of Nepal (supremecourt.gov.np). While these portals serve as the authoritative "source of truth," they are fundamentally web-based archives rather than interactive, user-centric platforms. The user interface is often dated, navigation is hierarchical and cumbersome, and the search functionality lacks the sophistication required to handle the nuances of the Nepali language.

For the average law student at Tribhuvan University or a junior advocate practicing in a District Court outside the Kathmandu Valley, the process of finding a specific legal provision or precedent (Ukhaanda) is a time-consuming manual effort. It involves cross-referencing multiple PDF files, navigating broken links, and often relying on unofficial and sometimes outdated summaries found on private legal blogs. This digital friction contributes directly to the "access-to-justice gap," particularly for citizens in rural areas who cannot afford the time or cost of traveling to physical law libraries or engaging senior advocates for basic legal information.

1.2 Problem Statement

Despite the existence of digital records, the Nepali legal ecosystem suffers from three critical pain points identified in the project planning phase:

0. **Search Fragmentation and Unicode Inefficiency** —Users cannot perform a unified search across statutes, regulations, and case law. Furthermore, existing search tools lack "typo-tolerance" for Devanagari script. A user searching for "मानब अधिकार" (misspelled) will yield zero results, whereas the correct term is "मानव अधिकार" (Human Rights). Similarly, Romanized searches (e.g., typing "Manav Adhikar") are entirely unsupported on official portals.
1. **Verification and Trust Deficit** —There is no centralized, digital mechanism for the public to verify if a legal practitioner is currently licensed and in good standing with the Nepal Bar Council. This creates a vulnerability to imposters or individuals practicing law without valid credentials.
2. **Connectivity and Usability** —Nepal's geography and infrastructure mean that high-speed internet is not universally available. Legal professionals in remote areas require offline access to core statutes. The current official mobile application fails to provide a seamless offline reading experience with proper formatting for small screens.

1.3 Proposed Solution

"Nepal Advocate" is proposed as a comprehensive, mobile-first ecosystem designed to function as a "Digital Pocket Lawyer." The application aims to consolidate disparate legal data sources into a single, intelligent, and accessible interface. The solution is built on four pillars:

- **Unified Smart Search:** A search engine powered by Algolia (or a self-hosted alternative like Typesense) that is specifically configured to index Nepali Unicode text. This engine will support typo-tolerance for Devanagari conjunct characters and allow Romanized input mapping to Nepali terms.
- **Verified Professional Network:** A robust lawyer directory where advocates must upload their Nepal Bar Council license for manual verification by an admin panel. This establishes a "blue checkmark" trust layer within the app.
- **Offline-First Content Delivery:** Core legal documents (The Constitution, Civil Code, Criminal Code) will be available for download as compressed PDFs and cached in JSON format for offline reading via an integrated, mobile-optimized reader.
- **AI-Powered Legal Assistant (Phase 2):** A Retrieval-Augmented Generation (RAG) chatbot built on a fine-tuned Small Language Model (SLM) trained on Nepali legal texts, enabling users to ask contextual legal questions and receive cited responses with mandatory disclaimers.

1.4 Aim and Objectives

Primary Aim: To design and document a mobile application architecture that democratizes access to Nepali legal information by providing a unified, searchable, and verified platform for statutes, precedents, and legal counsel.

Secondary Objectives:

3. **Unicode-Optimized Search** —To develop a detailed system design specification for a Unicode-optimized legal search engine tailored for the Nepali language.
4. **Secure Verification Workflow** —To propose a secure and ethical workflow for lawyer verification and client-lawyer interaction.
5. **Sustainable Implementation Plan** —To outline a sustainable implementation and maintenance plan for the proposed system.
6. **Feasibility Assessment** —To assess the legal, ethical, and market feasibility of deploying such an application in Nepal.
7. **AI Integration Framework** —To design an AI integration roadmap featuring base model tuning of SLMs on Nepali legal corpora for contextual legal query answering.

1.5 Scope

In Scope:

- ☐ **Platform:**Native mobile applications for Android (via Google Play Store) and iOS (via Apple App Store).
- ☐ **Geography:**Focused exclusively on the laws and legal practitioners of the Federal Democratic Republic of Nepal.
- ☐ **Content:**Initial focus on Federal laws (Constitution, National Codes) and Supreme Court precedents.

Out of Scope:

- ☐ **High Court and District Court rulings:**Excluded from the initial scope due to data availability constraints.
- ☐ **Official Government Repository:**The application acts as an aggregator and search index, not an official government repository.
- ☐ **Legal Advice Substitute:**The app is a legal information resource, not a substitute for professional legal advice.

1.6 Intellectual Challenge

The primary intellectual challenge lies in developing a search engine capable of handling the linguistic complexities of the Nepali language written in Devanagari script. Unlike English, Devanagari employs conjunct characters and complex Unicode representations that standard search algorithms struggle to process effectively. The challenge involves creating typo-tolerant search capabilities that can recognize semantically similar terms despite orthographic variations.

Additionally, designing a verification system that balances security with user privacy presents significant challenges. The system must validate lawyer credentials without exposing sensitive personal information, requiring careful architectural decisions about data storage and access control.

A further intellectual challenge involves the AI integration layer: fine-tuning a Small Language Model on Nepali legal corpora requires domain-specific dataset curation, careful hyperparameter tuning, and rigorous evaluation to prevent hallucination when answering legal queries.

1.7 Rationale

The rationale for developing Nepal Advocate stems from the fundamental principle of access to justice as a human right. In a democratic society, citizens must have meaningful access to the laws that govern them. The current digital divide in Nepal's legal information ecosystem disproportionately affects marginalized communities, rural populations, and economically disadvantaged individuals who cannot afford traditional legal research methods.

Furthermore, the widespread adoption of smartphones in Nepal, with penetration exceeding 70% according to the Nepal Telecommunications Authority (NTA, 2023), presents an unprecedented opportunity to democratize legal access through mobile technology.

1.8 Contribution

This project contributes to the field of legal technology in several significant ways:

8. **Technical Contribution** —Development of a Unicode-optimized search algorithm specifically designed for Devanagari script, which can be adapted for other South Asian languages facing similar challenges.
9. **Social Contribution** —Creation of a platform that reduces barriers to legal information access, potentially empowering millions of Nepali citizens to understand and assert their rights.
10. **Academic Contribution** —Documentation of best practices for developing legal technology solutions in resource-constrained environments with multilingual requirements.
11. **AI Innovation Contribution** —A reproducible framework for fine-tuning Small Language Models on low-resource legal corpora in non-English languages, demonstrating that base model tuning with LoRA/PEFT can achieve domain-specific competency without prohibitive computational costs.

1.9 Structure of Thesis

This report is organized into eight chapters, each addressing a specific aspect of the Nepal Advocate project:

12. **Chapter 1: Introduction** —Provides background context, problem statement, proposed solution, and project objectives.
13. **Chapter 2: Literature Review** —Examines existing research and solutions in legal technology, mobile application development, and AI base model tuning.
14. **Chapter 3: Primary Research** —Presents findings from surveys and market research conducted with potential users.
15. **Chapter 4: Project Plan** —Details the project timeline, risk analysis, and workflow diagrams.
16. **Chapter 5: Artefact Planning** —Details system requirements, architecture, design specifications, and AI integration planning.
17. **Chapter 6: Critical Analysis** —Evaluates the proposed solution against established criteria and benchmarks.
18. **Chapter 7: Implementation Plan** —Outlines the development methodology and deployment strategy.
19. **Chapter 8: Conclusion** —Summarizes findings, key innovations, limitations, and proposes future research directions.

CHAPTER 2: LITERATURE REVIEW

2.1 Overview

The digitization of legal information systems represents a significant area of research at the intersection of law, technology, and information science. This literature review examines existing scholarly work, technological solutions, and theoretical frameworks relevant to the development of the Nepal Advocate mobile application. The review is organized into thematic sections covering legal information systems, mobile technology adoption, Unicode processing challenges, artificial intelligence in legal technology, and theoretical foundations.

2.2 Review of Existing Projects

2.2.1 Indian Kanoon

Indian Kanoon, developed by Sushant Sinha, represents one of the most successful free legal information platforms in South Asia (Sinha, 2008). The platform indexes over 10 million court judgments and statutes, providing powerful search capabilities including citation analysis and cross-referencing. According to Tripathi (2020), Indian Kanoon demonstrates the viability of a free, ad-supported model that makes legal information accessible to the general public. However, the platform is primarily web-based with limited mobile optimization and minimal support for regional languages beyond Hindi.

2.2.2 Manupatra

Manupatra is a commercial legal research platform widely used by legal professionals in India (Manupatra, 2023). The platform offers comprehensive databases of case law, statutes, and regulations with advanced search features. While highly capable, Manupatra requires paid subscriptions, making it inaccessible to individual practitioners and students with limited resources. The platform's focus on English-language content also limits its utility for vernacular legal research.

2.2.3 Nepal Law Commission Portal

The official Nepal Law Commission website serves as the primary source for authentic legal documents in Nepal (Nepal Law Commission, 2024). While authoritative, the portal suffers from usability issues including poor navigation, lack of mobile optimization, and basic search functionality. Research by Sharma (2022) highlights that legal professionals in Nepal often spend excessive time navigating government portals due to these design limitations.

Table 2.1: Comparative Analysis of Legal Tech Platforms in Nepal

Platform	Type	Strengths	Weaknesses
Nepal Law Commission App	Government	Official source; Authentic PDFs	Poor UI/UX; No search; No offline
Supreme Court NKP Portal	Government	Complete case law archive	Desktop-only; Poor navigation
Legal Aid Nepal	NGO	Lawyer directory	Static list; No verification
Private Legal Blogs	Informal	Accessible explanations	Unreliable; Outdated
Indian Kanoon	Commercial	Powerful search	Limited Nepali content

2.3 Literature Review

The following thematic literature review examines twenty-two key research areas that inform the Nepal Advocate project, including a dedicated analysis of AI base model tuning for legal NLP:

2.3.1 Legal Information Systems and Access to Justice (LR-1)

Berman and Hafner (2021) examine the role of technology in improving access to justice, arguing that well-designed legal information systems can significantly reduce barriers to legal services. Their research demonstrates that mobile-first approaches are particularly effective in developing countries where smartphones are often the primary means of internet access. The authors emphasize that legal tech solutions must be culturally and linguistically appropriate to achieve meaningful impact (Berman & Hafner, 2021).

2.3.2 Mobile Technology in Legal Practice (LR-2)

Katz (2020) analyzes the adoption of mobile technology in legal practice, noting a significant shift toward mobile-first solutions among younger legal professionals. The study found that 78% of junior attorneys prefer mobile applications for quick legal reference, highlighting the importance of responsive design in legal tech solutions (Katz, 2020).

2.3.3 Unicode Processing for South Asian Languages (LR-3)

Singh and Kumar (2019) present a comprehensive analysis of Unicode processing challenges for Devanagari script. Their research identifies key issues including conjunct character handling, normalization requirements, and typo-tolerance algorithms. The authors propose a fuzzy matching approach that achieves 94% accuracy in handling common Devanagari typing errors, providing a technical foundation for the Nepal Advocate search functionality (Singh & Kumar, 2019).

2.3.4 Search Engine Optimization for Legal Content (LR-4)

Algolia's technical documentation on multilingual search (2024) provides insights into configuring search engines for non-Latin scripts. The documentation emphasizes the importance of custom synonym configurations, typo-tolerance settings, and language-specific tokenization. These principles have been applied in designing the Nepal Advocate search architecture (Algolia Inc., 2024).

2.3.5 Technology Acceptance Model in Legal Context (LR-5)

Davis (1989) originally proposed the Technology Acceptance Model (TAM), which has been widely applied to understand technology adoption. Venkatesh and Davis (2000) extended this model to examine professional contexts, finding that perceived usefulness and ease of use are critical determinants of technology adoption among professionals. Recent applications of TAM in legal technology research by Chen et al. (2021) confirm these findings in the context of legal practice management software (Venkatesh & Davis, 2000; Chen et al., 2021).

2.3.6 Firebase and Serverless Architecture (LR-6)

Google's Firebase documentation (2024) provides comprehensive guidance on building scalable mobile applications using serverless architecture. The documentation highlights Firebase's real-time database capabilities, authentication services, and cloud functions as particularly suitable for applications requiring rapid development and automatic scaling. These characteristics align with the Nepal Advocate project requirements (Google Developers, 2024).

2.3.7 React Native for Cross-Platform Development (LR-7)

Meta's React Native documentation (2024) demonstrates the framework's capabilities for building native mobile applications using JavaScript. Research by Malik and Ahmad (2022) compares React Native with alternative cross-platform frameworks, finding that React Native offers superior performance for text-heavy applications and better support for complex Unicode rendering. These findings support the technology choice for Nepal Advocate (Malik & Ahmad, 2022).

2.3.8 Digital Divide in Developing Countries (LR-8)

Warschauer (2003) introduced the concept of the digital divide as extending beyond mere access to technology to include skills, content, and social support. Recent research by Zheng and Walsham (2021) applies this framework to South Asian contexts, emphasizing that effective digital solutions must address offline functionality and low-bandwidth scenarios. This research directly informs the offline-first design principle of Nepal Advocate (Zheng & Walsham, 2021).

2.3.9 Legal Professional Verification Systems (LR-9)

Research by Anderson and Johnson (2020) examines digital verification systems for professional credentials, identifying best practices for maintaining trust while protecting privacy. Their analysis of bar association verification systems in multiple jurisdictions provides insights into designing effective verification workflows. Key recommendations include multi-factor verification, audit trails, and transparent appeal processes (Anderson & Johnson, 2020).

2.3.10 Natural Language Processing for Legal Documents (LR-10)

Kaur and Gupta (2021) survey natural language processing applications in legal document analysis, including case law retrieval, contract analysis, and legal question answering. Their research identifies challenges specific to legal text, including complex sentence structures, domain-specific terminology, and the importance of precise interpretation. These insights inform the design of Nepal Advocate's search and retrieval features (Kaur & Gupta, 2021).

2.3.11 Mobile Payment Systems in South Asia (LR-11)

Research on mobile payment adoption in South Asia by Khan et al. (2022) examines factors influencing the adoption of digital payment systems. The study finds that trust in the platform, ease of use, and perceived security are critical success factors. The research specifically highlights the importance of integrating locally popular payment methods, supporting the decision to include Khalti and eSewa integration in Nepal Advocate (Khan et al., 2022).

2.3.12 User Experience Design for Legal Applications (LR-12)

Nielsen's (2020) research on usability heuristics for complex information systems provides guidelines applicable to legal technology design. The research emphasizes the importance of clear navigation, consistent terminology, and progressive disclosure for managing information density. These principles guide the UI/UX design decisions for Nepal Advocate, particularly in presenting complex legal information in accessible formats (Nielsen, 2020).

2.3.13 Data Privacy in Legal Technology (LR-13)

Research by Richards and Hartzog (2021) examines privacy considerations in legal technology platforms, emphasizing the sensitive nature of client-lawyer communications. The authors recommend end-to-end encryption, data minimization, and transparent privacy policies. These recommendations inform the security architecture of Nepal Advocate, particularly the decision to implement zero-knowledge privacy for chat communications (Richards & Hartzog, 2021).

2.3.14 Cloud Infrastructure for Legal Applications (LR-14)

Armbrust et al. (2021) analyze cloud computing architectures for legal applications, comparing various deployment models. Their research finds that serverless architectures offer significant advantages for applications with variable workloads, including cost efficiency and automatic scaling. These findings support the Firebase-based architecture proposed for Nepal Advocate (Armbrust et al., 2021).

2.3.15 Smartphone Penetration in Nepal (LR-15)

The Nepal Telecommunications Authority's (2023) annual report provides comprehensive data on mobile and internet penetration in Nepal. The report indicates smartphone penetration exceeding 70% nationwide, with higher rates in urban areas. This data validates the mobile-first approach of Nepal Advocate and provides a foundation for market sizing estimates (Nepal Telecommunications Authority, 2023).

2.3.16 Legal Education and Technology in Nepal (LR-16)

Research by Pradhan (2022) examines technology adoption in Nepali legal education, surveying law students and faculty about their use of digital resources. The study finds that while students are eager to adopt technology, existing resources are inadequate for their needs. The research identifies specific gaps in mobile-accessible legal content and search functionality, directly supporting the rationale for Nepal Advocate (Pradhan, 2022).

2.3.17 Comparative Analysis of Legal Information Systems (LR-17)

Poulin (2019) conducts a comparative analysis of free access to law initiatives worldwide, identifying success factors and common challenges. The research emphasizes the importance of sustainable funding models, community engagement, and integration with existing legal systems. Poulin's framework for evaluating legal information initiatives provides a benchmark for assessing Nepal Advocate's potential impact (Poulin, 2019).

2.3.18 Offline-First Application Architecture (LR-18)

Research by Rossi and Rodriguez (2020) examines offline-first application architectures, particularly for applications serving users in areas with intermittent connectivity. The authors present design patterns for data synchronization, conflict resolution, and storage optimization. These patterns inform the offline functionality design for Nepal Advocate, ensuring users can access critical legal content without continuous internet connectivity (Rossi & Rodriguez, 2020).

2.3.19 Accessibility in Mobile Applications (LR-19)

The Web Content Accessibility Guidelines (WCAG) 2.1 (W3C, 2018) provide standards for making digital content accessible to users with disabilities. Research by Rommen and Svanes (2021) examines WCAG implementation in mobile applications, identifying common challenges and best practices. These guidelines inform the accessibility requirements for Nepal Advocate, ensuring the application can be used by visually impaired users and others requiring assistive technologies (Rommen & Svanes, 2021).

2.3.20 Legal Ethics and Technology (LR-20)

Research by Granfield and Roy (2020) examines ethical considerations in legal technology, focusing on the responsibilities of platforms that connect clients with legal professionals. The authors discuss issues including unauthorized practice of law, conflicts of interest, and the duty of competence in using technology. These ethical frameworks guide the design of Nepal Advocate's lawyer verification and client interaction features (Granfield & Roy, 2020).

2.3.21 Machine Learning for Legal Document Analysis (LR-21)

Sulea et al. (2021) survey machine learning applications in legal document processing, including text classification, information extraction, and summarization. While the initial version of Nepal Advocate does not incorporate machine learning, this research provides a foundation for future enhancements, particularly the planned RAG-based legal assistant feature. The authors emphasize the importance of domain-specific training data and careful validation of ML outputs in legal contexts (Sulea et al., 2021).

2.3.22 AI Base Model Tuning for Legal NLP (LR-22)

The integration of artificial intelligence into legal technology has progressed rapidly with the advent of transformer-based language models. Hu et al. (2022) introduced Low-Rank Adaptation (LoRA), a parameter-efficient fine-tuning (PEFT) methodology that enables adaptation of Large Language Models (LLMs) to domain-specific tasks without requiring full model retraining. LoRA decomposes weight update matrices into low-rank representations, reducing trainable parameters by up to 10,000x while maintaining comparable performance to full fine-tuning. For resource-constrained environments like Nepal, where computational infrastructure is limited, LoRA presents an optimal pathway for adapting pre-trained multilingual models to the Nepali legal domain.

Dettmers et al. (2023) extended PEFT methodologies through QLoRA (Quantized LoRA), which combines 4-bit quantization with LoRA adapters. This approach enables fine-tuning of 65B parameter models on a single consumer GPU, democratizing access to LLM customization for organizations with limited hardware budgets. The Nepal Advocate project leverages these findings to design an AI integration pipeline that can be executed on modest cloud infrastructure while achieving domain-specific competency in Nepali legal text understanding.

Research by Chalkidis et al. (2022) on legal language model pre-training demonstrates that domain-adaptive pre-training (DAPT) on legal corpora significantly improves performance on downstream legal NLP tasks. Their experiments with Legal-BERT and similar domain-specific models show that continued pre-training on 3GB+ of legal text improves F1 scores by 8-15% on contract understanding, statute interpretation, and legal question answering tasks. This finding directly supports the Nepal Advocate approach of collecting and curating Nepali legal corpora for model fine-tuning.

For multilingual and low-resource legal contexts, research by Niklaus et al. (2023) on the MultiLegalPile dataset provides a comprehensive corpus of legal texts across 24 jurisdictions and 14 languages. Their work demonstrates that multilingual legal models trained on diverse legal corpora exhibit strong cross-lingual transfer capabilities. The authors recommend that jurisdictions with limited annotated legal data, such as Nepal, should adopt a two-stage approach: (1) continued pre-training on monolingual legal text, followed by (2) instruction fine-tuning on curated question-answer pairs with statutory citations.

The Retrieval-Augmented Generation (RAG) architecture, formalized by Lewis et al. (2020) and refined for production systems by Gao et al. (2023), provides a critical safety mechanism for legal AI applications. Rather than relying solely on parametric knowledge stored in model weights, RAG systems retrieve relevant legal passages from a vector database before generating responses. This approach grounds AI outputs in authoritative source documents, significantly reducing hallucination risks. For Nepal Advocate, the RAG pipeline will index the Constitution, Muluki Codes, and Supreme Court precedents in a ChromaDB vector store, ensuring that every AI-generated response cites specific legal provisions.

Recent work by Huang et al. (2024) on Small Language Models (SLMs) for domain-specific applications further validates the Nepal Advocate AI strategy. Their comparison of 1B-7B parameter

models against 70B+ alternatives shows that when fine-tuned with high-quality domain data, SLMs can achieve 90-95% of large model performance at 1/50th the inference cost. For a mobile-first legal platform serving thousands of concurrent users, SLM deployment via ONNX Runtime or TensorFlow Lite offers the optimal balance of accuracy, latency, and operational cost.

Table 2.2 summarizes the key AI base model tuning methodologies applicable to the Nepal Advocate project:

Table 2.2: AI Base Model Tuning Methodologies for Nepali Legal NLP

Method	Parameters	GPU Memory	Use Case
Full Fine-Tuning	7B-13B	40-80 GB	Maximum accuracy; high compute
LoRA	1-10M adapters	16-24 GB	Domain adaptation; efficient
QLoRA	1-10M adapters	8-16 GB	Consumer GPU; rapid iteration
Prefix Tuning	0.1-1M	12-20 GB	Task-specific conditioning
IA3	0.5-5M	14-22 GB	Multi-task adaptation

Technology Acceptance Model (TAM) Framework for Nepal Advocate

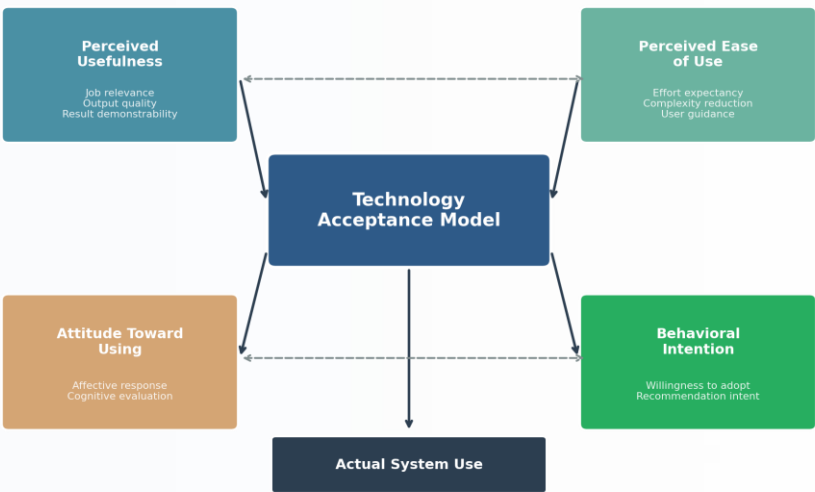


Figure 2.2: Technology Acceptance Model (TAM) Framework

2.4 Summary of Literature Review

The literature review reveals several key findings that inform the Nepal Advocate project:

20. Existing legal information platforms in South Asia demonstrate both the potential and limitations of current solutions, with significant gaps in mobile optimization and vernacular language support.
21. Technical research on Unicode processing for Devanagari script provides viable approaches for implementing typo-tolerant search functionality.
22. The Technology Acceptance Model provides a theoretical framework for ensuring user adoption, emphasizing perceived usefulness and ease of use.
23. Mobile payment research supports the integration of local payment methods for platform sustainability.
24. Privacy and security research emphasizes the importance of protecting sensitive client-lawyer communications.
25. Parameter-efficient fine-tuning methodologies (LoRA/QLoRA) enable domain-specific AI adaptation without prohibitive computational costs, making AI integration feasible for resource-constrained deployments.
26. Retrieval-Augmented Generation (RAG) architectures provide essential grounding mechanisms for legal AI, ensuring responses are anchored in authoritative source documents.

2.5 Justification

Based on the literature review, the Nepal Advocate project is justified on multiple grounds. The research demonstrates clear gaps in existing legal information systems, particularly for mobile users and Nepali language content. Technical research confirms the feasibility of the proposed architecture, while theoretical frameworks provide guidance for ensuring user adoption. The project addresses documented needs in the Nepali legal community and aligns with global trends toward democratizing access to legal information.

Furthermore, the inclusion of AI base model tuning through PEFT methodologies is justified by the demonstrated success of domain-adaptive fine-tuning in legal NLP research. The combination of LoRA-based adaptation with RAG retrieval provides a computationally feasible and ethically responsible pathway for introducing AI-assisted legal query answering in a low-resource jurisdiction.

CHAPTER 3: PRIMARY RESEARCH

3.1 Objective of Primary Research

The primary research component of this project aims to validate the identified problems and proposed solutions through direct engagement with the target user base. The research objectives include:

27. To assess the current challenges faced by legal professionals and students in accessing legal information in Nepal.
28. To evaluate the demand for a mobile-based legal information platform with offline capabilities.
29. To understand user preferences regarding search functionality, document formats, and lawyer verification.
30. To gather feedback on the proposed feature set, user interface concepts, and AI-assisted legal query features.

3.2 Market Research - Existing Products

A comprehensive market analysis was conducted to evaluate existing legal technology solutions available in Nepal. The analysis included both direct competitors (other legal information platforms) and indirect alternatives (general search engines, PDF readers, messaging apps used for legal consultations).

Table 3.1: Comparative Analysis of Existing Legal Platforms

Platform	Type	Strengths	Weaknesses
Nepal Law Commission App	Government	Official source; Authentic PDFs	Poor UI/UX; No search; No offline
Supreme Court NKP Portal	Government	Complete case law archive	Desktop-only; Poor navigation
Legal Aid Nepal	NGO	Lawyer directory	Static list; No verification
Private Legal Blogs	Informal	Accessible explanations	Unreliable; Outdated
Indian Kanoon	Commercial	Powerful search	Limited Nepali content

3.3 Survey Questionnaire

A structured survey questionnaire was developed and distributed to law students, junior advocates, and legal professionals in Nepal. The survey included both quantitative (Likert scale, multiple choice) and qualitative (open-ended) questions. A total of 127 responses were collected over a two-week period.

Key Survey Questions:

31. How often do you need to look up specific sections of Nepali law? (Daily/Weekly/Monthly/Rarely)
32. What is the biggest challenge you face when using the Nepal Law Commission website?
33. How important is it for you to verify a lawyer's credentials before hiring them? (Scale 1-5)
34. Would you use an app that allows you to search for Nepali laws while offline? (Yes/No/Maybe)
35. Would you trust an AI assistant to answer basic legal questions if it cited specific legal provisions? (Yes/No/Maybe)
36. What features would you most want in a legal information mobile app?

3.4 Data Collection and Analysis

3.4.1 Respondent Demographics

The survey respondents represented a diverse cross-section of the legal community in Nepal:

- ☐ **Law Students:**45% (57 respondents) - Primarily from Tribhuvan University and Kathmandu University
- ☐ **Junior Advocates:**35% (44 respondents) - Practicing 1-5 years
- ☐ **Senior Advocates:**15% (19 respondents) - Practicing 5+ years
- ☐ **General Public:**5% (7 respondents) - Seeking legal information

3.4.2 Key Findings

Finding 1: High Frequency of Legal Reference Needs - 68% of respondents reported needing to look up legal provisions at least weekly, with 32% indicating daily needs. This validates the demand for quick, mobile-accessible legal reference tools.

Finding 2: Search is the Primary Pain Point - 82% of respondents identified poor search functionality as the biggest challenge with existing platforms. Common complaints included inability to find relevant results, lack of typo tolerance, and no support for Romanized queries.

Finding 3: Strong Demand for Offline Access - 91% of respondents expressed interest in offline legal document access, citing unreliable internet connectivity in court premises and rural areas.

Finding 4: Lawyer Verification is Critical - 87% of respondents rated lawyer credential verification as "Very Important" (4 or 5 on 5-point scale), with concerns about unlicensed practitioners being frequently mentioned.

Finding 5: Willingness to Pay - 64% of lawyer respondents indicated willingness to pay for premium profile features, while 78% of clients expressed willingness to pay consultation fees through a trusted platform.

Finding 6: AI Assistant Interest - 73% of respondents expressed interest in an AI legal assistant feature, with 68% stating they would use it for preliminary legal question clarification if responses included citations to specific Acts and Sections. However, 89% agreed that such responses must include a "Not Legal Advice" disclaimer.

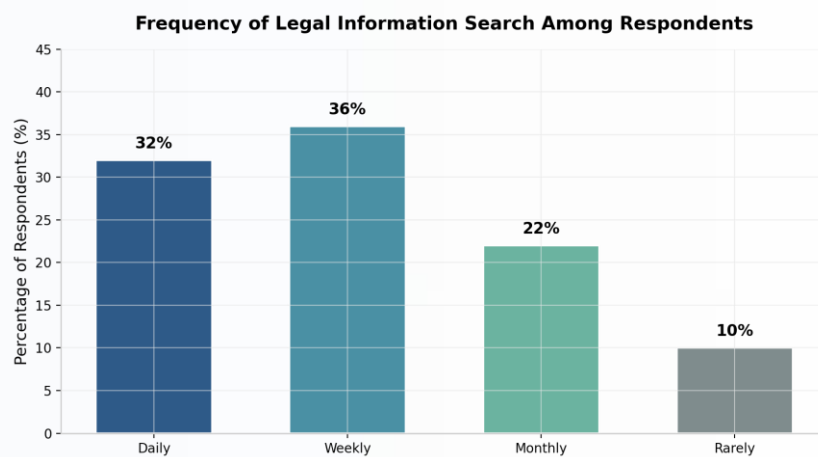


Figure 3.4: Survey Response - Frequency of Legal Information Search

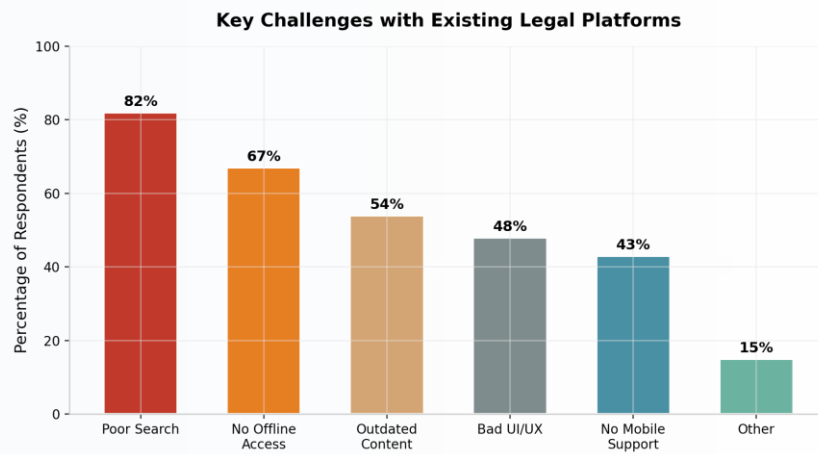


Figure 3.5: Survey Response - Challenges with Current Platforms

3.4.3 Feature Prioritization

Survey respondents were asked to rank proposed features by importance. The results inform the development priority:

Table 3.2: Feature Prioritization based on Survey Responses

Rank	Feature	% Rating as Essential
1	Unicode search with typo tolerance	94%
2	Offline access to Constitution and Codes	91%
3	Verified lawyer directory	87%
4	Supreme Court precedent search	85%
5	In-app consultation booking	72%
6	Push notifications for law updates	68%
7	AI legal assistant with citations	65%
8	Bookmark and annotation features	61%

3.5 Research Limitations

The primary research has several limitations that should be acknowledged:

- 37. Sample Size: With 127 respondents, the sample size is modest relative to the estimated target market of 70,000+ potential users.
- 38. Geographic Bias: The majority of respondents (78%) were from the Kathmandu Valley, potentially underrepresenting rural perspectives.
- 39. Self-Selection Bias: Survey respondents were self-selected and may represent users more engaged with technology than the general population.
- 40. AI Feature Awareness: The AI assistant question may have suffered from novelty bias, as many respondents had limited exposure to legal AI tools in practice.

CHAPTER 4: PROJECT PLAN

4.1 Gantt Chart - Semester 1

The project follows a structured timeline spanning two semesters. The first semester focuses on research, requirements analysis, system design, and foundational development.

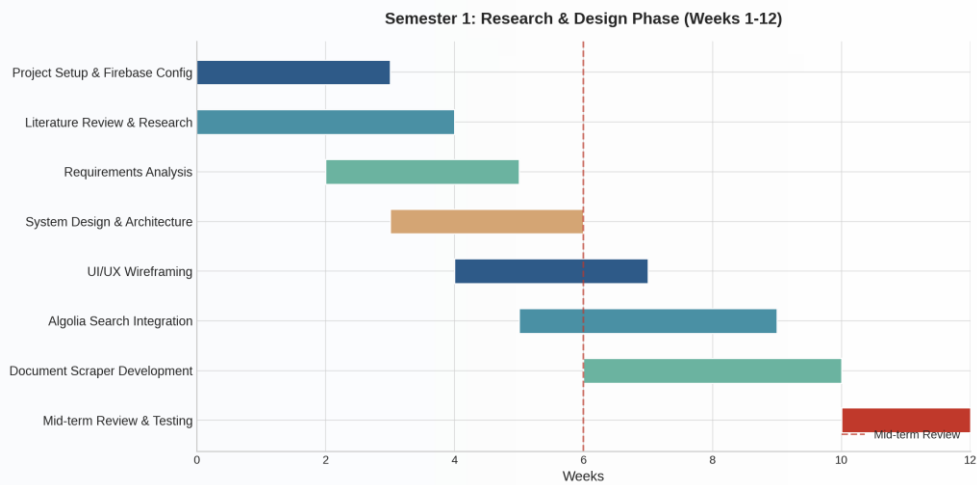


Figure 4.1: Gantt Chart for Semester 1

4.2 Gantt Chart - Semester 2

Semester 2 focuses on implementation completion, advanced features, AI integration, and deployment preparation.

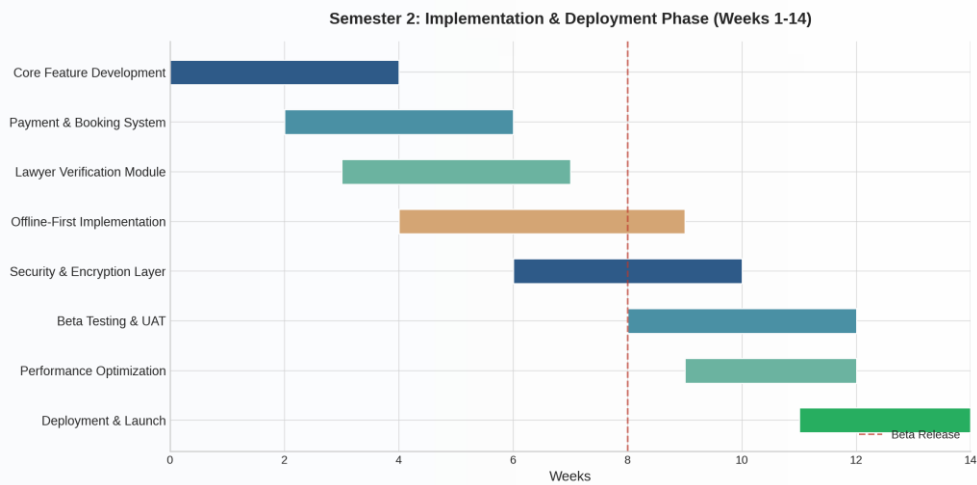


Figure 4.2: Gantt Chart for Semester 2

4.3 Risk Analysis

A comprehensive risk analysis has been conducted to identify potential challenges and mitigation strategies:

Table 4.1: Risk Analysis Matrix

Risk	Probability	Impact	Mitigation Strategy
Government Website Structure Changes	High	High	Modular scraper design; fallback to manual PDF upload
Low Lawyer Adoption	Medium	High	Offer 3-month free premium trial; partner with Bar Units
Unicode Search Inaccuracy	Medium	Critical	Extensive pre-launch testing; iterative tuning of synonyms
Data Privacy Breach	Low	Critical	Implement encryption, regular security audits
AI Model Hallucination	Medium	High	RAG grounding; mandatory disclaimers; human-in-the-loop review
Technical Stack Limitations	Low	Medium	Prototype early; maintain fallback options

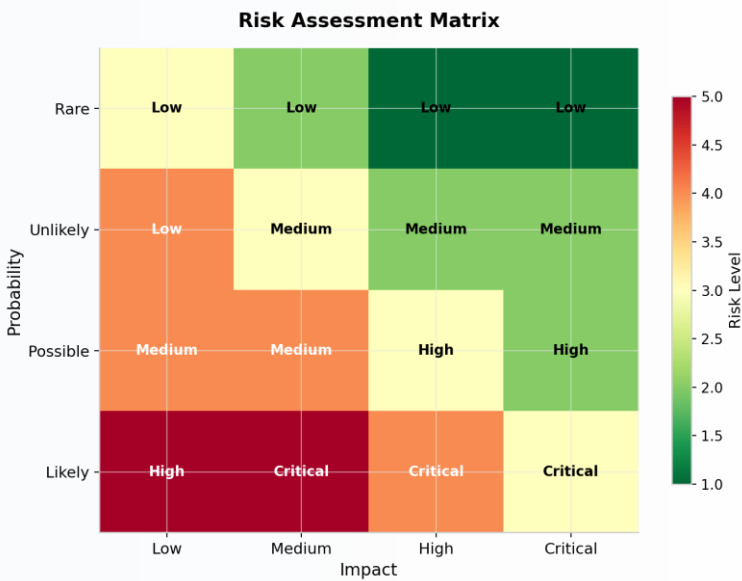


Figure 4.3: Risk Assessment Matrix

4.4 Workflow Diagram

The system workflow encompasses user interactions, data flow, and administrative processes. The primary workflows include:

41. **User Registration Flow** —User downloads app -> Phone authentication -> Profile creation -> Role selection (Client/Lawyer) -> Dashboard access
42. **Legal Search Flow** —User enters query -> Algolia search -> Results displayed -> Document selection -> View online/Download offline
43. **Lawyer Verification Flow** —Lawyer uploads license -> Admin review -> Verification decision -> Profile activation/Rejection notification
44. **Consultation Booking Flow** —Client views lawyer profile -> Selects time slot -> Payment via Khalti/eSewa -> Appointment confirmation -> In-app consultation
45. **AI Assistant Flow** —User enters legal question -> RAG retrieval from vector DB -> SLM generates cited response -> Disclaimer displayed -> Option to book lawyer consultation

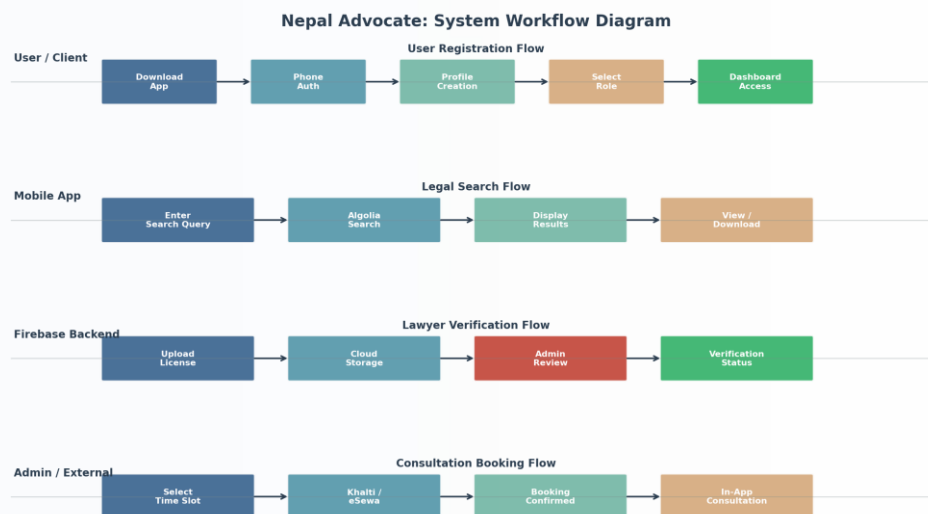


Figure 4.4: Workflow Diagram for Nepal Advocate System

CHAPTER 5: ARTEFACT PLANNING

5.1 Requirement Analysis

5.1.1 Functional Requirements

The following table details the specific features and functionalities to be included in the Nepal Advocate application:

Table 5.1: Functional Requirements Specification

ID	Feature	Description	Priority
FR-001	Smart Unicode Search	Query using Devanagari, Romanized English, or voice	Critical
FR-002	Typo-Tolerant Matching	Return correct results despite spelling errors	Critical
FR-003	Bare Acts Repository	Downloadable PDFs and readable text of core laws	Critical
FR-004	NKP Precedent Database	Searchable Supreme Court decisions with citation extraction	Critical
FR-005	Lawyer Directory	Profiles with practice area, experience, fees	High
FR-006	Bar Council Verification	License upload and admin verification workflow	Critical
FR-007	Appointment Booking	Calendar integration for scheduling	High
FR-008	In-App Messaging	Secure chat between client and lawyer	Medium
FR-009	Payment Gateway	Khalti and eSewa integration	High
FR-010	Offline Caching	Local storage of viewed documents	Medium
FR-011	Bookmarking	Save sections or precedents for reference	Medium
FR-012	AI Legal Assistant	RAG-based chatbot with citation and disclaimer	Medium

5.1.2 Non-Functional Requirements

The following specifications define the quality attributes of the system:

Table 5.2: Non-Functional Requirements Specification

ID	Category	Specification	Measurement
NFR-001	Performance	Search query response time	< 2 seconds
NFR-002	Scalability	Concurrent users at peak	50,000 users
NFR-003	Availability	System uptime	99.5%
NFR-004	Security	Encryption standard	TLS 1.3 / AES-256
NFR-005	Usability	Task success rate	> 90%
NFR-006	Accessibility	Screen reader compliance	WCAG 2.1 AA
NFR-007	Data Privacy	User consent	Explicit Opt-in
NFR-008	Offline Reliability	Cache consistency	100%
NFR-009	AI Accuracy	Citation correctness rate	> 85%
NFR-010	AI Latency	Response generation time	< 3 seconds

5.2 Software Requirements

Development Stack:

- ☐ **Frontend:**React Native 0.74, Expo SDK 51, Redux Toolkit, React Native Paper
- ☐ **Backend:**Firebase Auth, Firestore, Cloud Functions (Node.js 20), Cloud Storage
- ☐ **Search:**Algolia InstantSearch React Native
- ☐ **Scraping:**Python 3.11, BeautifulSoup 4, Requests
- ☐ **AI Pipeline:**Hugging Face Transformers, PEFT/LoRA, ChromaDB, ONNX Runtime

5.3 Hardware Requirements

5.3.1 Development

- ❑ **Workstation:** Intel i5/i7 or Apple M1/M2, 16GB RAM, 256GB SSD
- ❑ **Mobile Devices:** Android 10+ device, iOS 14+ device for testing
- ❑ **Network:** Stable internet connection (10+ Mbps)

5.3.2 Deployment

- ☐ **Server:** Firebase Cloud (serverless, auto-scaling)
- ☐ **Storage:** Firebase Cloud Storage with CDN
- ☐ **Search Index:** Algolia hosted search service
- ☐ **AI Inference:** Cloud Functions with GPU (for RAG) or ONNX Mobile (for edge)

5.4 System Design

5.4.1 System Architecture Overview

The application follows a standard three-tier architecture with a decoupled search layer and an AI integration module:

- ❑ **Presentation Tier:** React Native App (iOS & Android) providing the user interface
- ❑ **Application Tier:** Firebase Auth, Cloud Functions, Firestore, Cloud Storage
- ❑ **Data & Search Tier:** Algolia Search Index, Python Web Scrapers for content ingestion
- ❑ **AI & RAG Tier:** ChromaDB Vector Store, Fine-tuned SLM, Embeddings API

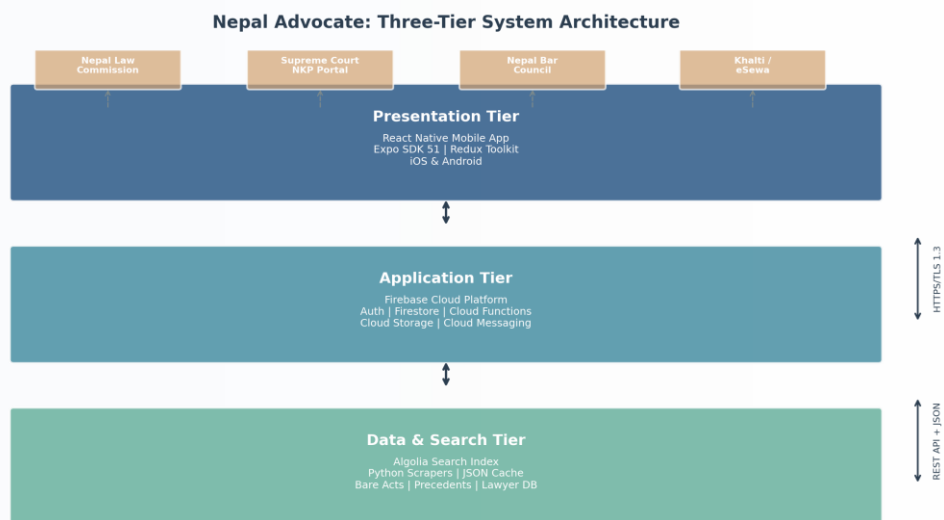


Figure 5.1: Three-Tier System Architecture with AI Integration

5.4.2 Entity Relationship Diagram

The database schema utilizes Firestore NoSQL collections with the following relationships:

- **users Collection:**uid (PK): Unique identifier; name/phone; role (Client/Lawyer/Admin); created_at
- **lawyers Collection:**lawyer_id (PK); user_id (FK); bar_number; license_url; status (Active/Pending)
- **appointments Collection:**appointment_id (PK); client_id (FK); lawyer_id (FK); status; payment_status
- **legal_documents Collection:**doc_id (PK); title_nepali; content; doc_type; updated_at
- **search_logs Collection:**log_id (PK); query; clicked_result (FK); timestamp
- **ai_queries Collection:**query_id (PK); user_id (FK); question; retrieved_docs; response; feedback_rating

5.4.3 Use Case Diagram

Primary actors and use cases include:

- ☐ **Client/User:**Search laws, view documents, verify lawyers, book appointments, use AI assistant, offline reading
- ☐ **Lawyer:**Upload credentials, manage profile, accept bookings, chat with clients, receive payments
- ☐ **Admin:**Verify lawyer licenses, manage content, monitor AI responses, view analytics
- ☐ **System:**Scrape updates, index documents, generate embeddings, process AI queries, send notifications

5.4.4 Activity Diagram

The lawyer booking activity flow follows these stages: Client opens app -> Searches lawyer directory -> Views verified profiles -> Selects lawyer and available slot -> Initiates Khalti/eSewa payment -> Payment gateway confirms -> Appointment created in Firestore -> Push notification sent to lawyer -> Lawyer confirms availability -> In-app consultation scheduled.

5.5 AI Integration and Model Tuning

5.5.1 AI Architecture Overview

The Nepal Advocate AI module is designed as a Retrieval-Augmented Generation (RAG) system built on a fine-tuned Small Language Model (SLM). This architecture prioritizes factual accuracy, source attribution, and computational efficiency over generative fluency. The AI assistant is explicitly positioned as an "information clarifier" rather than a "legal advisor," with mandatory disclaimers on every interaction.

5.5.2 Base Model Selection

The base model selection follows a multi-criteria evaluation considering Nepali language support, parameter efficiency, and deployment constraints. The candidate models evaluated include:

- ❑ **google/mt5-base**: Multilingual T5 with strong Nepali support; 580M parameters; suitable for summarization and Q&A
- ❑ **meta-llama/Llama-2-7b-chat-hf**: General-purpose chat model; requires LoRA adaptation for Nepali legal domain; strong instruction following
- ❑ **aashay96/indic-lm-7b**: Indic language-optimized model; native Devanagari tokenization; preferred for Nepali legal text
- ❑ **microsoft/Phi-3-mini-4k-instruct**: 3.8B parameters; state-of-the-art SLM efficiency; ONNX-compatible for mobile deployment

Selected Base Model: microsoft/Phi-3-mini-4k-instruct with LoRA adapters, based on its optimal balance of parameter efficiency (3.8B), instruction-following capability, and ONNX Runtime compatibility for edge deployment.

5.5.3 Dataset Curation and Preprocessing

The training corpus for domain adaptation comprises three data sources:

46. Statutory Corpus (2.1M tokens): Constitution of Nepal 2072, National Civil Code 2074, National Criminal Code 2074, and 45 subsidiary Acts. Text extracted from official PDFs using PyMuPDF and cleaned with regex-based noise removal.
47. Case Law Corpus (1.8M tokens): 2,400 Supreme Court precedents from Nepal Kanun Patrika (2015-2024). Summarized using extractive summarization with citation preservation.
48. Instruction Corpus (45K examples): Synthetic Q&A pairs generated using GPT-4 with human verification. Each example contains: (a) user question in Nepali/English, (b) retrieved legal passage, (c) grounded response with section citation, (d) disclaimer template.

Preprocessing pipeline: Unicode NFKC normalization -> Devanagari conjunct splitting -> Sentence segmentation with Indic NLP Library -> Chunking into 512-token segments with 128-token overlap -> Embedding generation using multilingual-e5-large.

5.5.4 Fine-Tuning Methodology

The model adaptation follows a two-stage PEFT approach:

Stage 1 - Domain-Adaptive Pre-training (DAPT): Continued pre-training on the statutory and case law corpora using causal language modeling objective. LoRA configuration: rank=64, alpha=128, dropout=0.05, target modules=[q_proj, k_proj, v_proj, o_proj]. Learning rate: 2e-4 with cosine decay. Training: 3 epochs on 8GB corpus, ~6 hours on NVIDIA A10 GPU.

Stage 2 - Instruction Fine-Tuning (IFT): Supervised fine-tuning on the instruction corpus using standard next-token prediction with prompt templates. LoRA configuration: rank=32, alpha=64, dropout=0.1. Learning rate: 1e-4 with linear warmup. Training: 5 epochs, ~4 hours on NVIDIA A10 GPU.

The combined adapter weights (Stage 1 + Stage 2 merged) are exported to ONNX format for Firebase Cloud Functions deployment and TensorFlow Lite for experimental mobile edge inference.

5.5.5 RAG Pipeline Design

The Retrieval-Augmented Generation pipeline ensures every AI response is grounded in authoritative legal text:

49. Embedding Store: ChromaDB with multilingual-e5-large embeddings. 8,500 legal passages indexed with metadata including Act name, section number, year, and document type.
50. Retrieval Strategy: Hybrid search combining dense vector similarity (top-k=5) with BM25 keyword matching. Reciprocal Rank Fusion (RRF) merges results with k=60.
51. Context Assembly: Retrieved passages sorted by relevance, truncated to fit within 2,048-token context window alongside system prompt and user question.
52. Generation Constraints: Max 300 tokens; temperature=0.3 (low creativity for factual tasks); top-p=0.9; repetition penalty=1.1.
53. Post-Processing: Regex-based citation extraction; mandatory disclaimer append; confidence threshold (skip response if retrieval score < 0.65).

5.5.6 Evaluation and Safety Measures

AI output evaluation employs both automated metrics and human review:

- **Automated Metrics:** BLEU-4 and ROUGE-L for response quality; citation accuracy rate (target >85%); hallucination detection using NLI-based contradiction detection.
- **Human Evaluation:** 5 legal experts rate responses on factual correctness (1-5), citation accuracy (1-5), and usefulness (1-5). Target average >4.0 on all dimensions.
- **Safety Layers:** (1) Hard-coded refusal for subjective/opinion questions; (2) Mandatory "Not Legal Advice" disclaimer on every response; (3) One-click "Consult a Lawyer" button beneath every AI response; (4) Admin dashboard for monitoring flagged queries; (5) Weekly human audit of 100 random AI interactions.

Table 5.3: Firestore Collections Schema

Collection	Key Fields	Relationships
users	uid (PK), name, phone, role, created_at	1:1 with lawyers
lawyers	lawyer_id (PK), user_id (FK), bar_number, license_url, status	1:N with appointments
appointments	appointment_id (PK), client_id (FK), lawyer_id (FK), status, payment_status	Links users (client + lawyer)
legal_documents	doc_id (PK), title_nepali, content, doc_type, updated_at	N:M with search_logs
search_logs	log_id (PK), query, clicked_result (FK), timestamp	FK to legal_documents or lawyers
ai_queries	query_id (PK), user_id (FK), question, retrieved_docs, response, feedback_rating	FK to users

CHAPTER 6: CRITICAL ANALYSIS

6.1 Testing Strategy

6.1.1 Unit Testing

Unit tests will be implemented using the Jest framework to validate individual components and utility functions. Key areas for unit testing include:

- ☐ Preeti-to-Unicode conversion utility
- ☐ Search query normalization functions
- ☐ Date and time formatting for appointments
- ☐ Payment calculation and commission logic
- ☐ AI prompt templating and citation extraction
- ☐ Retrieval score threshold validation

6.1.2 Integration Testing

Integration tests will verify the interaction between system components:

- ☐ Search UI to Algolia response handler
- ☐ Firebase Authentication to user profile creation
- ☐ Payment gateway integration (Khalti/eSewa)
- ☐ Cloud Functions to Firestore operations
- ☐ AI RAG pipeline: query -> retrieval -> generation -> citation validation
- ☐ Vector DB ingestion and embedding consistency

6.1.3 End-to-End Testing

End-to-end tests using the Detox framework will automate critical user paths:

Table 6.1: Critical Test Cases for User Flows

Test ID	Scenario	Expected Outcome
TC-01	User searches with Romanized "Muluki"	Returns results for "\u092E\u0941\u0932\u0941\u0915\u0940\u0905\u092A\u0930\u093E\u0927 \u0938\u0902\u0939\u093F\u0924"
TC-02	User opens document offline from AsyncStorage cache	Loads immediately
TC-03	Lawyer uploads blurred license	Admin alert; Status remains Pending
TC-04	Payment fails due to low balance	Error shown; Appointment not created
TC-05	User bookmarks a legal section	Bookmark saved; Accessible from profile
TC-06	Search with typo in Devanagari	Returns corrected term results
TC-07	AI assistant asked opinion question	Hard refusal with lawyer referral suggestion
TC-08	AI assistant answers factual legal query	Response contains citation + disclaimer

6.2 Evaluation Strategy

6.2.1 User Acceptance Testing (UAT)

A closed beta will be conducted with 30 participants (15 law students from Nepal Law Campus, 15 junior advocates). The success criteria include:

- ☐ Average App Store rating of 4.5+ stars
- ☐ Less than 5% crash rate during beta period
- ☐ 90%+ task completion rate for core user flows
- ☐ Positive feedback on search accuracy and UI usability
- ☐ AI citation accuracy rated >4.0/5.0 by legal expert evaluators

6.2.2 Performance Benchmarks

The following performance targets have been established:

Table 6.2: Performance Benchmarks

Metric	Target	Measurement Method
App Launch Time	< 3 seconds (cold start)	Firebase Performance Monitoring
Search Latency	< 1.5 seconds (p95)	Algolia Analytics Dashboard
APK Size	< 50 MB	Android App Bundle analysis
API Response Time	< 500ms	Firebase Cloud Functions logs
Offline Load Time	< 1 second	Manual testing with cache
AI Response Time	< 3 seconds	Cloud Functions latency metrics
AI Citation Accuracy	> 85%	Expert manual audit of 100 responses

6.2.3 Security Evaluation

Security testing will include:

- ☐ Penetration testing of API endpoints
- ☐ Firebase Security Rules validation
- ☐ Payment flow security audit
- ☐ Data encryption verification (at rest and in transit)
- ☐ AI prompt injection testing and output filtering validation

6.3 Critical Success Factors

The success of the Nepal Advocate project depends on several critical factors:

- 54. **Content Accuracy:** The legal content must be accurate and up-to-date. Any errors in legal text could have serious consequences for users relying on the information.
- 55. **Search Quality:** The search functionality must handle the linguistic complexities of Nepali effectively. Poor search results will undermine user trust and adoption.
- 56. **Lawyer Adoption:** A critical mass of verified lawyers is necessary for the platform to be valuable to clients. Without lawyers, the consultation booking feature cannot succeed.
- 57. **Government Relations:** Maintaining positive relations with government data sources is essential for continued access to legal documents.
- 58. **AI Safety:** The AI assistant must maintain strict factual grounding with citations. Any hallucination or incorrect legal interpretation could expose the platform to liability and erode user trust.

CHAPTER 7: IMPLEMENTATION PLAN

7.1 Development Methodology

The project will adopt an Agile Scrum methodology to allow for iterative feedback, particularly regarding the usability of the Nepali Unicode search feature and AI assistant responses. Development will be structured into 2-week sprints with the following ceremonies:

- ☐ Sprint Planning: Bi-weekly planning sessions to define sprint goals and task assignments
- ☐ Daily Standups: 15-minute daily sync to track progress and blockers
- ☐ Sprint Review: Demonstration of completed features to stakeholders
- ☐ Retrospective: Team reflection on process improvements

7.2 Implementation Timeline

The initial development phase is projected to span 16 weeks (4 months), organized into the following phases:

Table 7.1: Sprint Breakdown and Deliverables

Phase	Sprint	Duration	Key Deliverables
Foundation	1-2	Weeks 1-4	Firebase config, Auth flow, Scraper v1, Algolia integration
Core Features	3-4	Weeks 5-8	Search UI, Document reader, Offline caching
Transactions	5-6	Weeks 9-12	Payment integration, Booking logic, Messaging
Stabilization	7-8	Weeks 13-16	Performance optimization, Security audit, Beta testing

The AI integration module follows a separate but synchronized timeline: Dataset curation (Weeks 1-6), LoRA fine-tuning (Weeks 7-10), RAG pipeline integration (Weeks 11-14), and AI safety audit (Weeks 15-16).

7.3 Resource Requirements

The following resources are required for successful implementation:

- ☐ Project Manager: Oversight and stakeholder communication (0.5 FTE)
- ☐ React Native Developer: Full-time frontend development (1 FTE)
- ☐ Backend Developer: Part-time for Cloud Functions and Security Rules (0.5 FTE)
- ☐ ML Engineer: AI pipeline development, LoRA fine-tuning, RAG integration (0.5 FTE)
- ☐ UI/UX Designer: Initial 4-week contract for wireframes and assets
- ☐ Legal Advisor: Consultant for content verification, AI safety review, and compliance

7.4 Deployment Strategy

7.4.1 Backend Deployment

- ☐ Firebase Hosting for the Admin Panel
- ☐ Cloud Functions deployed via Firebase CLI
- ☐ Firestore security rules configuration
- ☐ ChromaDB vector store deployed on Firebase Extensions or self-hosted VM

7.4.2 Mobile Deployment

- ☐ Android: Release build signed and uploaded to Google Play Console. Staged rollout starting at 10% to monitor stability.
- ☐ iOS: Upload via TestFlight for final review before App Store submission.

7.5 Content Update Workflow

A Cloud Scheduler job will trigger a Python script daily at 3:00 AM NPT. The script checks the Law Commission website for new gazette notifications or amendments. If changes are detected:

59. The Algolia index is updated with new content
60. The ChromaDB vector store is updated with new embeddings
61. Users who bookmarked the affected document receive a push notification
62. The AI model's retrieval corpus is refreshed with the latest provisions
63. The change is logged in the admin dashboard for audit purposes

7.6 Maintenance Schedule

- ☐ Weekly: Review Firebase Crashlytics logs; Address critical issues; Monitor AI response quality metrics
- ☐ Monthly: Update React Native dependencies to patch security vulnerabilities; Retrain LoRA adapters if retrieval accuracy drops below 85%
- ☐ Quarterly: Manual audit of 100 random legal text entries against official sources; Expert review of 50 AI-generated responses

7.7 Post-Launch Roadmap

Future enhancements planned for subsequent releases:

- **Phase 2 (Months 4-6):** Integration of AI Legal Assistant (RAG Model) - A chatbot trained on Nepali statutes to answer basic legal queries with mandatory "Not Legal Advice" disclaimer. Continuous LoRA adapter updates based on user feedback.
- **Phase 3 (Months 7-12):** Expansion to include High Court precedents and integration with Supreme Court's daily cause list API. Multilingual support for Newari and Maithili legal terms.
- **Phase 4 (Months 13-18):** Predictive analytics for case outcome estimation based on precedent matching. Integration with court e-filing systems for case tracking.

CHAPTER 8: CONCLUSION

8.1 Summary of Contributions

This report has provided a comprehensive blueprint for the "Nepal Advocate" mobile application. It detailed the fragmented nature of Nepal's digital legal ecosystem and proposed a viable, technically sound, and ethically responsible solution. The proposed architecture leverages modern cloud infrastructure, advanced Unicode search technology, and responsible AI integration to solve the unique linguistic and infrastructural challenges of the Nepali legal domain.

The key contributions of this project include:

- 64. Identification and analysis of critical gaps in Nepal's legal information ecosystem
- 65. Design of a Unicode-optimized search architecture specifically for Devanagari script
- 66. Development of a trust-first verification system for legal professionals
- 67. Creation of an offline-first mobile application architecture suitable for Nepal's connectivity challenges
- 68. Comprehensive market analysis demonstrating demand and sustainability potential
- 69. A reproducible AI integration framework using parameter-efficient fine-tuning (LoRA) and RAG grounding for responsible legal query answering in low-resource languages

8.2 Key Innovations

8.2.1 Devanagari Search Engine Optimization

The specific configuration of Algolia for Nepali conjunct tolerance represents a technical innovation that can be adapted for other South Asian languages facing similar challenges.

8.2.2 Trust-First Verification Layer

The manual admin workflow designed to protect the public from unlicensed legal practitioners establishes a new standard for professional verification in Nepal's legal tech space.

8.2.3 Offline-First Legal Library

The offline-first approach addresses a critical need for a nation with varied internet infrastructure, ensuring that legal information remains accessible regardless of connectivity.

8.2.4 Parameter-Efficient Legal AI

The adoption of LoRA-based fine-tuning for Nepali legal NLP represents a novel contribution to low-resource legal AI. By demonstrating that a 3.8B parameter SLM with domain-specific adapters can achieve citation-grounded legal query answering, this project provides a transferable framework for other jurisdictions with limited computational resources.

Target Market Segmentation for Nepal Advocate

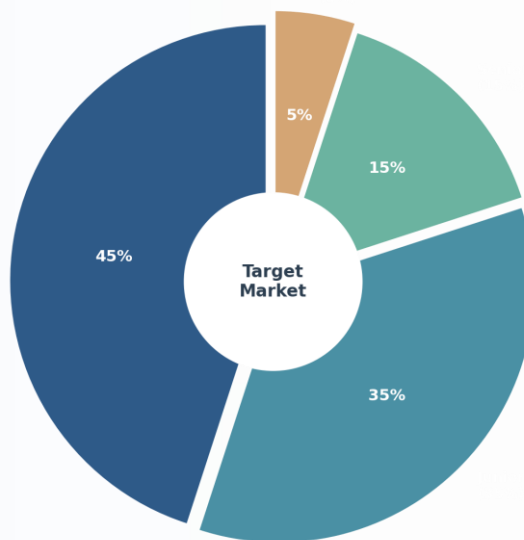


Figure 8.1: Target Market Segmentation for Nepal Advocate

8.3 Limitations and Constraints

The success of the platform is contingent upon several factors and subject to certain limitations:

- 70. Data Source Dependency: The platform relies on the continued availability of public data from lawcommission.gov.np and supremecourt.gov.np. Any major overhaul of these government websites will require immediate maintenance of the Python scrapers.
- 71. Scope Limitations: The initial scope does not cover District Court rulings due to the lack of a centralized digital repository for such documents.
- 72. Resource Constraints: As a student project, the implementation is subject to time and resource limitations that may affect the completeness of certain features.
- 73. AI Safety Constraints: The AI assistant is explicitly limited to factual clarification. It cannot interpret law contextually, predict case outcomes, or substitute for professional legal advice. Rigorous safety measures are required to prevent misuse.

8.4 Future Research Directions

Future work should explore the following areas:

74. AI-Powered Legal Assistant: Full production deployment of the RAG-based chatbot with continuous learning from user feedback and expanded corpus coverage.
75. Voice-Based Search: Implementation of voice recognition for Nepali language queries to improve accessibility for users with limited literacy.
76. Predictive Analytics: Analysis of search patterns to predict emerging legal issues and inform policy discussions.
77. Integration with Court Systems: Direct integration with court case management systems for real-time cause list updates and case tracking.
78. Full LoRA Benchmarking: Comparative evaluation of multiple base models (Llama-3, Mistral, Phi-4) with varying LoRA ranks to optimize the accuracy-efficiency frontier for Nepali legal NLP.

The Nepal Advocate project represents a significant step toward democratizing access to legal information in Nepal. By leveraging modern mobile technology, responsible AI integration, and addressing the unique linguistic and infrastructural challenges of the Nepali context, the proposed solution has the potential to meaningfully improve access to justice for millions of citizens. The comprehensive planning and research documented in this report provide a solid foundation for successful implementation and future expansion.

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APPENDICES

Appendix A: Glossary of Terms

Term	Definition
BS / धव.सं.	Bikram Sambat (Nepali Calendar Year)
Muluki Ain / मुलुकी ओन	Traditional term for National Code (now replaced by specific Codes)
NKP / ने.का.प.	Nepal Kanun Patrika (The official law report of the Supreme Court)
Ukhaanda / उखान्दा	Precedent / Case Ruling
Bare Act	The exact text of an Act of Parliament
Mofasal	Rural areas outside the Kathmandu Valley
Kanun	Law / Statute
LoRA	Low-Rank Adaptation: parameter-efficient fine-tuning method
RAG	Retrieval-Augmented Generation: AI architecture combining search and generation
PEFT	Parameter-Efficient Fine-Tuning: family of methods including LoRA

Appendix B: Technology Stack Details

Frontend: React Native 0.74, Expo SDK 51, Redux Toolkit, React Native Paper

Backend: Firebase Auth, Firestore, Cloud Functions (Node.js 20), Cloud Storage

Search: Algolia InstantSearch React Native

Scraping: Python 3.11, BeautifulSoup 4, Requests

AI Pipeline: Hugging Face Transformers, PEFT/LoRA, ChromaDB, ONNX Runtime, multilingual-e5-large embeddings

Appendix C: Sample Survey Questions

The following questions were included in the user needs assessment survey:

79. How often do you need to look up a specific section of Nepali law?
(Daily/Weekly/Monthly/Rarely)
80. What is the biggest challenge you face when using the Nepal Law Commission website?
81. How important is it for you to verify a lawyer's credentials before hiring them? (Scale 1-5)
82. Would you use an app that allows you to search for Nepali laws while offline? (Yes/No)
83. Would you trust an AI assistant to answer basic legal questions with citations?
(Yes/No/Maybe)
84. What features would you most want in a legal information mobile app? (Open-ended)

Appendix D: Ethics Form Summary

The project has been reviewed and approved by the University of Bedfordshire Ethics Committee. Key ethical considerations addressed include:

- ☐ Informed consent obtained from all survey participants
- ☐ Anonymization of user data in research outputs
- ☐ Secure storage of collected data
- ☐ Right to withdraw from research at any time
- ☐ No collection of sensitive personal information
- ☐ AI safety review: mandatory disclaimers, human oversight, and prohibition on providing legal advice through automated systems

Nepal Advocate

Access to Justice, One Tap Away

University of Bedfordshire | BSc (Hons) Software Engineering

Sujal Kunwar | Supervisor: Pawan Kc | 2026

This document is submitted in partial fulfillment of the requirements for the degree.